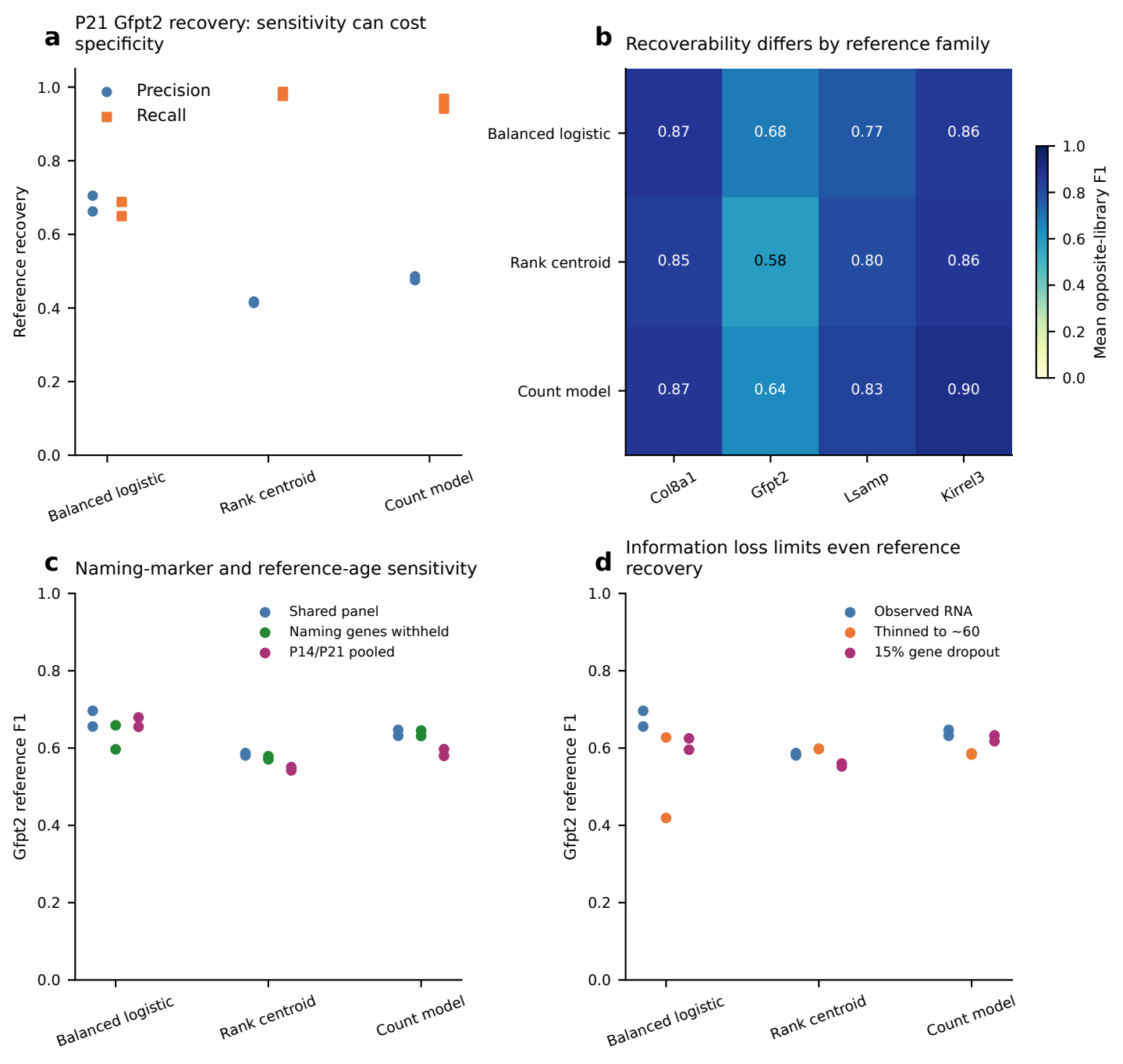
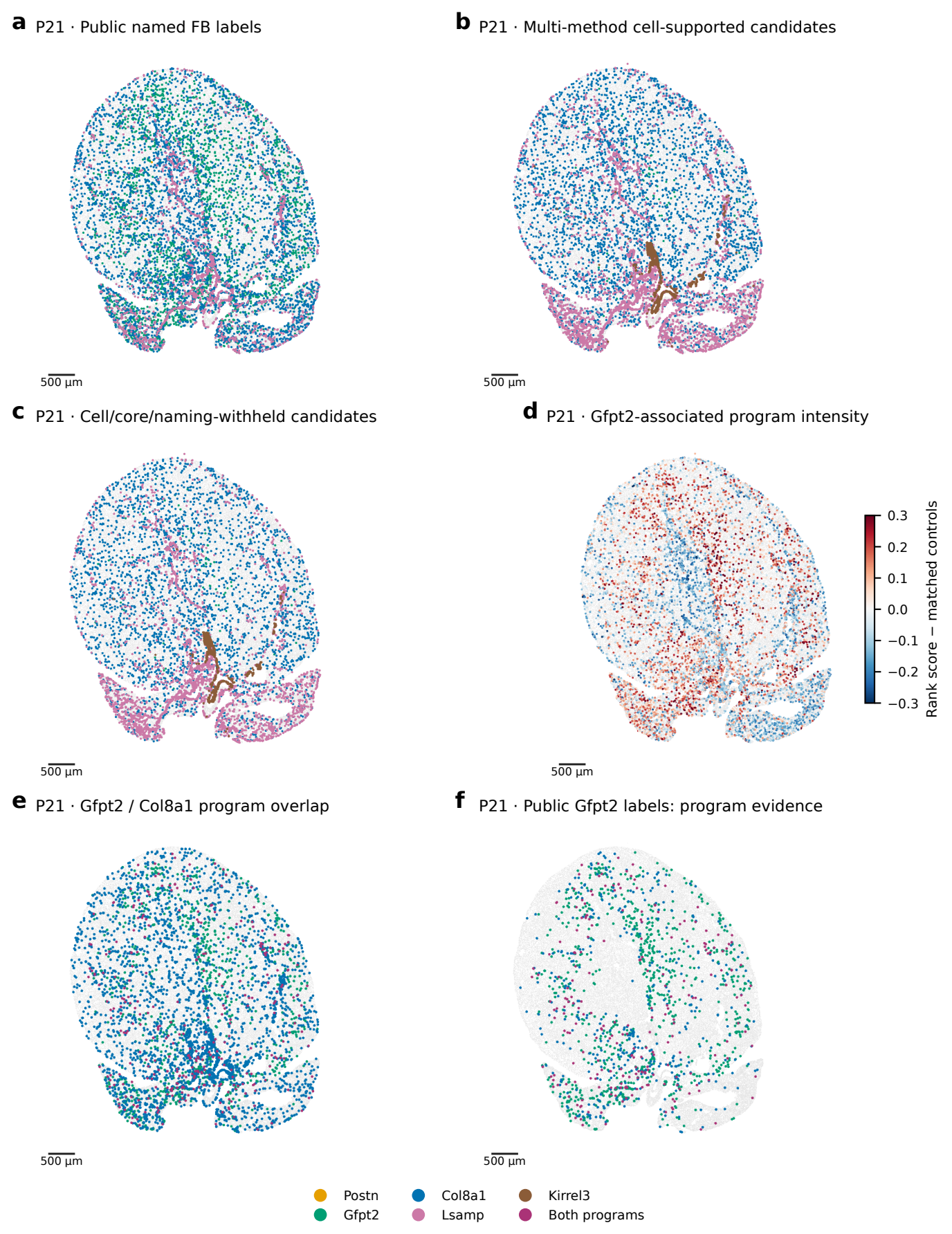
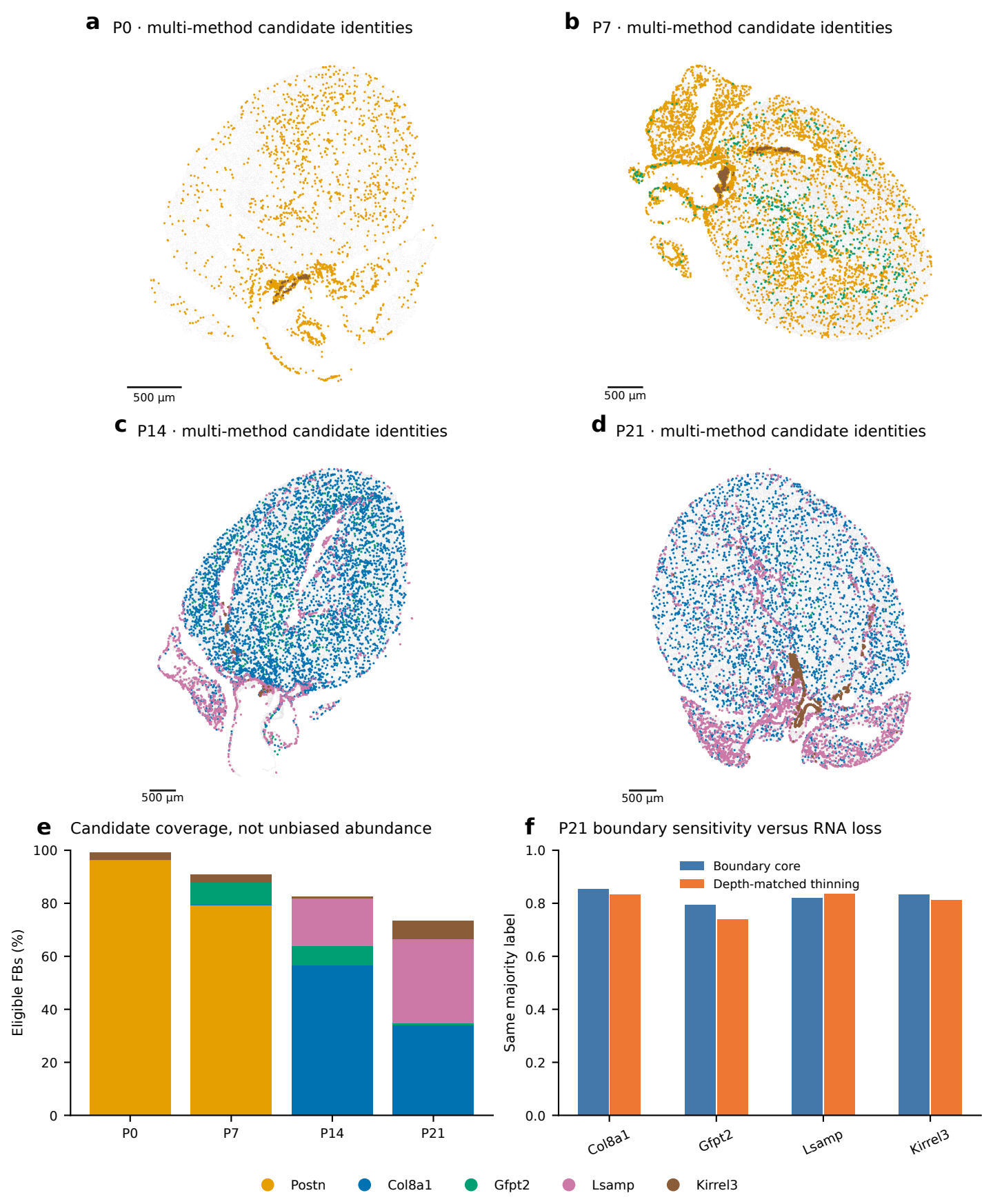
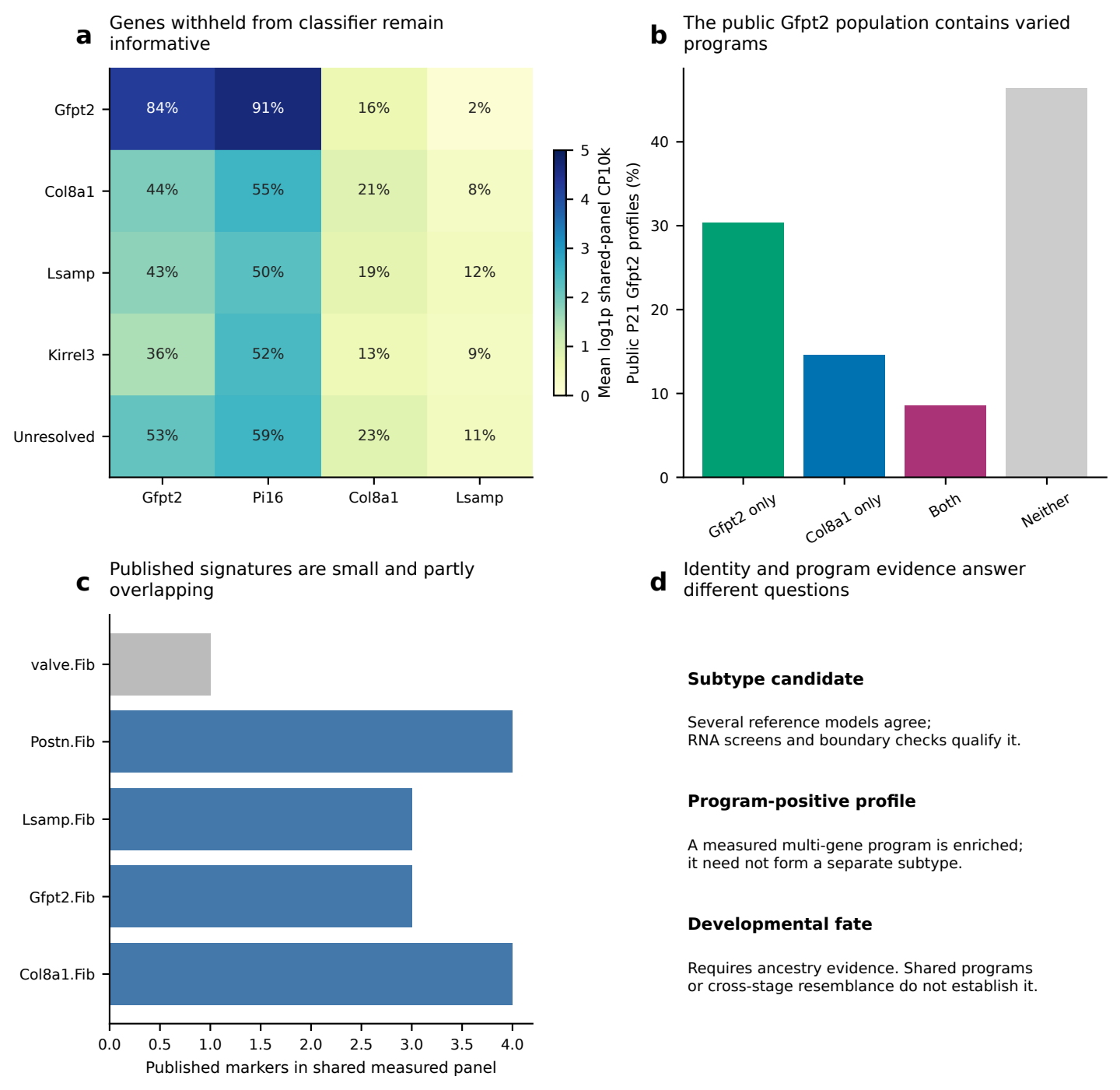

a
P21 Gfpt2 recovery: sensitivity can cost specificity
Precision
Recall
Reference recovery
1.0
0.8
0.6
0.4
0.2
0.0
Balanced logistic
Rank centroid
Count model
b
Recoverability differs by reference family
Balanced logistic
Rank centroid
Count model
0.87
0.68
0.77
0.86
0.85
0.58
0.80
0.86
0.87
0.64
0.83
0.90
Col8a1
Gfpt2
Lsamp
Kirrel3
Mean opposite-library F1
1.0
0.8
0.6
0.4
0.2
0.0
c
Naming-marker and reference-age sensitivity
Shared panel
Naming genes withheld
P14/P21 pooled
Gfpt2 reference F1
1.0
0.8
0.6
0.4
0.2
0.0
Balanced logistic
Rank centroid
Count model
d
Information loss limits even reference recovery
Observed RNA
Thinned to ~60
15% gene dropout
Gfpt2 reference F1
1.0
0.8
0.6
0.4
0.2
0.0
Balanced logistic
Rank centroid
Count model


a
P21 · Public named FB labels
500 µm
b
P21 · Multi-method cell-supported candidates
500 µm
c
P21 · Cell/core/naming-withheld candidates
500 µm
d
P21 · Gfpt2-associated program intensity
500 µm
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Rank score − matched controls
e
P21 · Gfpt2 / Col8a1 program overlap
500 µm
f
P21 · Public Gfpt2 labels: program evidence
500 µm
Postn
Gfpt2
Col8a1
Lsamp
Kirrel3
Both programs


a
P0 · multi-method candidate identities
500 µm
b
P7 · multi-method candidate identities
500 µm
c
P14 · multi-method candidate identities
500 µm
d
P21 · multi-method candidate identities
500 µm
e
Candidate coverage, not unbiased abundance
Eligible FBs (%)
100
80
60
40
20
0
P0
P7
P14
P21
f
P21 boundary sensitivity versus RNA loss
Same majority label
1.0
0.8
0.6
0.4
0.2
0.0
Boundary core
Depth-matched thinning
Col8a1
Gfpt2
Lsamp
Kirrel3
Postn
Col8a1
Gfpt2
Lsamp
Kirrel3


a
Genes withheld from classifier remain informative
Gfpt2
Col8a1
Lsamp
Kirrel3
Unresolved
84%
91%
16%
2%
44%
55%
21%
8%
43%
50%
19%
12%
36%
52%
13%
9%
53%
59%
23%
11%
Gfpt2
Pi16
Col8a1
Lsamp
Mean log1p shared-panel CP10k
0
1
2
3
4
5
b
The public Gfpt2 population contains varied programs
Public P21 Gfpt2 profiles (%)
0
10
20
30
40
Gfpt2 only
Col8a1 only
Both
Neither
c
Published signatures are small and partly overlapping
valve.Fib
Postn.Fib
Lsamp.Fib
Gfpt2.Fib
Col8a1.Fib
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Published markers in shared measured panel
d
Identity and program evidence answer different questions
Subtype candidate
Several reference models agree;
RNA screens and boundary checks qualify it.
Program-positive profile
A measured multi-gene program is enriched;
it need not form a separate subtype.
Developmental fate
Requires ancestry evidence. Shared programs
or cross-stage resemblance do not establish it.